

CAREERMATRI · CAREER DECISION BRIEF

Respondent

Mechanical Engineering · Master's · STANDARD · 2026-10-06

DECISION

The three to look at first

Look at this one first

Mechanical Design & Product Engineering

Interest High

Experience High

Core 6/6 confirmed

Grounds available to explain it

Compare alongside

CAE / Simulation / Computational Engineering

Interest High

Experience High

Core 3/9 confirmed

Comparing roles

Then this one

Thermal / Fluids / Energy Systems

Interest High

Experience High

Core 1/6 confirmed

Comparing roles

What matters most in this result

- CAE / Simulation / Computational Engineering has core 9 of 3 are confirmed, and Problem definition is empty.
- Thermal / Fluids / Energy Systems has core 6 of 1 are confirmed, and Load calculation is empty.
- The experience you recorded currently reaches Organisational value is confirmed.

Your most important evidence gap

Problem definition

CAE / Simulation / Computational Engineering
 · Framed, yourself, what had to be analysed and why

What to do next

Problem definition do one piece of work that fills that gap. Framed, yourself, what had to be analysed and why

Nothing here measures your ability or your odds of being hired. Only what your answers and recorded experience confirm is written here. The three roles below are an order to look in, not a ranking. When the differences are small, that order shifts easily.

DECISION

Comparing the roles to look at first

Where the cells diverge is what tells you what to do next.

Role	Work style	Interest	Experience	Learning intent	Core confirmed	Evidence depth	Current
Mechanical Design & Product Engineering	Further away	High	High	High	6 / 6	E4 Organisational value	Grounds available explain it
CAE / Simulation / Computational Engineering	Further away	High	High	High	3 / 9	E4 Organisational value	Comparin roles
Thermal / Fluids / Energy Systems	Further away	High	High	High	1 / 6	E4 Organisational value	Comparin roles
Worth keeping in view							
Automation / Robotics / Mechatronics	Further away	High	Low	Low	1 / 6	E4 Organisational value	Look at th
Engineering Data / Digital Engineering	Further away	High	Medium	Low	1 / 6	E4 Organisational value	Look at th

We do not merge the cells into one. Where interest is high but experience is empty, doing one small thing first beats applying, and comes first. To say that, the two cells have to be separated. All sixteen roles are in the appendix.

WHY

how what you learned is used in real work

This runs in one line from what you learned to what the organisation counts as a result.

Mechanical Design & Product Engineering

Settling on one buildable form while the requirements pull against each other

What you learned

Mechanics of materials Statics

This is what you met in class. Having taken a course is not the same as being able to do it

Real work

turn the requirements into dimensions and tolerances

set the structure with the loads in mind

compare the design options and choose one

Technical judgement

Thickness and geometry

Material

Safety margin

Output

Design options and drawings that go on to production

3D models and drawings

Design calculation sheet

What the organisation values

Performance

cost

manufacturability

Private company is the basis.

My evidence today

bracket weight reduction capstone · Organisational value

Evidence still needed

A drawing or 3D model taken through to completion

A comparison table of design options

A record of building it and correcting it

EVIDENCE

The evidence I have today

This looks at how far one experience has travelled, from activity through to repeatability. A lower rung does not mean something is lacking. It is the range confirmed by what you have recorded.

bracket weight reduction capstone

Activity	Confirmed	turned the requirements into dimensions and compared three design options
Judgement	Confirmed	checked the rib layout and thickness that cost less to machine · rib layout
Output	Confirmed	drawing · analysis report
Performance	Confirmed	15% margin against the allowable stress in test · compared against the allowable stress
Organisational value	Confirmed	adopted as the final design
Repeatability	Not yet	

Repeatability to get there Have you used the same method on another project

SolidWorks · Design and CAD

through to an output

Experience of use	Confirmed
Connected experience	bracket weight reduction capstone
where it was used	compare geometry options
what you judged	rib layout
what remains	drawing
what it was compared against	compared against the allowable stress

ANSYS Mechanical · CAE and structural analysis

through to an output

Experience of use	Confirmed
Connected experience	bracket weight reduction capstone
where it was used	compare design options
what you judged	rib layout
what remains	analysis report
what it was compared against	compared against the allowable stress

A tool name on its own only confirms the activity. count as a score.

ORGANIZATION VALUE

The same technical background creates different value in different organisations

The same knowledge produces a product in one place and a paper in another.

Private company

Output

Design options and drawings that go on to production

What counts as performance

Performance · cost · manufacturability · lead time

Government-funded research institute

Output

Design as a prototype and as a project deliverable

What counts as performance

Project goals met · technical level · transferability

University laboratory

Output

A prototype that shows the concept, and the reasoning behind its design

What counts as performance

Novelty · reproducibility · scholarly contribution

Engineering and technical consultancy

Output

A comparison of design options the client uses to decide

What counts as performance

Client requirements met · schedule · repeat business

The same Mechanical Design & Product Engineering knowledge reads as a different result in each organisation. Your fit does not change when the organisation does.

GAP

Not yet confirmed

This does not mean you cannot do it. It means that from what you have recorded, it is not confirmed.

How far each role's core areas are confirmed

Role	Confirmed so far	Partially confirmed	Not yet confirmed	Fill this first
CAE / Simulation / Computational Engineering	3 / 9	0	6	Problem definition · Assumptions set out
Thermal / Fluids / Energy Systems	1 / 6	0	5	Load calculation · System layout

ACTION

What to do, and when

What to do, and when

30 days

- Every core area is confirmed. Line your evidence up against three postings for the same role, one row at a time.
- pick out the methods from your degree research that carry into a role, and write them down

This plan comes from the gaps you have right now. Record more experience and this line changes.

90 days

- widen the evidence you have confirmed up to the level of outputs and performance
- try that method against an industry problem once
- set two roles side by side and see which reads your evidence better

This plan comes from the gaps you have right now. Record more experience and this line changes.

That is the end of the main report; what follows is an appendix for checking back.

APPENDIX · FOR CHECKING BACK

Material kept out of the main report

It holds all sixteen roles, the item trace, evidence coverage and how this was produced, and because the findings in the main report come from this material, nothing here changes them.

All sixteen role families

The main report carries only the three to look at first and the two to keep in view.

Role	Work style	Interest	Experience	Learning intent	Core confirmed	Current stage
Mechanical Design & Product Engineering	Further away	High	High	High	6 / 6	Grounds available to explain it
CAE / Simulation / Computational Engineering	Further away	High	High	High	3 / 9	Comparing roles
Thermal / Fluids / Energy Systems	Further away	High	High	High	1 / 6	Comparing roles
Automation / Robotics / Mechatronics	Further away	High	Low	Low	1 / 6	Look at this first
Engineering Data / Digital Engineering	Further away	High	Medium	Low	1 / 6	Look at this first
R&D / Advanced Development	Further away	High	Low	Low	2 / 8	Look at this first
Research Scientist / Academic Research	Further away	Medium	Medium	Low	0 / 7	Look at this first
Systems Engineering / Technical PM	Further away	Medium	Low	—	1 / 6	Look at this first
Materials / Tribology / Mechanical Failure	Further away	Medium	Low	—	0 / 5	Look at this first
Technical Consulting / Engineering Services	Further away	Medium	Low	—	3 / 6	Look at this first
Manufacturing / Production Engineering	Further away	Low	Low	Low	2 / 6	Set aside for now
Process / Industrialization Engineering	Further away	Low	Low	Low	0 / 6	Set aside for now
Quality / Reliability / Failure Analysis	Further away	Low	Low	Low	1 / 6	Set aside for now
Equipment / Maintenance / Facilities	Further away	Low	Low	Low	1 / 5	Set aside for now
Test / Verification / Validation	Further away	Low	Low	Low	1 / 6	Set aside for now
Applications / Field / Customer Engineering	Further away	Low	Low	—	1 / 5	Set aside for now

which questions produced this line

The main report drops the question numbers and gives the reason in plain words. When you want to trace it back we leave them here for you.

Role	Questions behind it
Mechanical Design & Product Engineering	Q1, Q17, Q41, Q50, Q53
CAE / Simulation / Computational Engineering	Q2, Q18, Q42, Q50
Thermal / Fluids / Energy Systems	Q10
Automation / Robotics / Mechatronics	Q9, Q24, Q45
Engineering Data / Digital Engineering	Q15, Q22, Q46, Q50

Work style in detail

This axis rests on two questions, so we put no number on it. Until it is validated we describe it only as close, mixed or further away.

Axis	Leaning	Questions
Independent ↔ Collaborative	Neither side	2
Challenge ↔ Stability	Neither side	2
Speed ↔ Quality	Neither side	2

whether a result produced once can be reused

produced a result once **Confirmed** bracket weight reduction capstone

used the same method again **Not yet**

carried it to another problem **Not yet**

handed it over for others to use **Not yet**

A degree does not raise it. It rises only from the projects and methods you recorded.

Full evidence coverage by role

Core, supporting and optional are all laid out. An empty optional area is never held against you.

Mechanical Design & Product Engineering

Core areas

Confirmed 6

Partial 0

Not yet 0

all 6

Confirmed so far

- ✓ Interpreting requirements
bracket weight reduction capstone
- ✓ Generating and comparing design options
bracket weight reduction capstone
- ✓ Judging geometry, dimensions and material
bracket weight reduction capstone
- ✓ CAD and drawing outputs
bracket weight reduction capstone
- ✓ Structural and functional review
bracket weight reduction capstone
- ✓ Validation or design review
bracket weight reduction capstone
- ✓ Cost Supporting
bracket weight reduction capstone
- ✓ Use of design tools Optional
bracket weight reduction capstone · Tools: SolidWorks

Not yet confirmed

- ☐ Tolerances Supporting
- ☐ Manufacturability Supporting
- ☐ Change control Supporting
- ☐ Interfaces Supporting
- ☐ Feedback from real build and test Supporting
- ☐ Industry standard Optional

CAE / Simulation / Computational Engineering

Core areas

Confirmed 3

Partial 0

Not yet 6

all 9

Confirmed so far

- ✓ Model creation
bracket weight reduction capstone
- ✓ Running the analysis
bracket weight reduction capstone
- ✓ Validation
bracket weight reduction capstone
- ✓ Convergence check Supporting
bracket weight reduction capstone
- ✓ Mesh independence Supporting
bracket weight reduction capstone
- ✓ Parametric study Supporting
bracket weight reduction capstone
- ✓ Automation and scripting Supporting
bracket weight reduction capstone
- ✓ Documenting the analysis Supporting
bracket weight reduction capstone
- ✓ Use of analysis tools Optional
bracket weight reduction capstone · Tools: ANSYS Mechanical

Not yet confirmed

- ☐ Problem definition
- ☐ Assumptions set out
- ☐ Boundary conditions and loads
- ☐ Material and model parameters
- ☐ Interpreting the result
- ☐ Link to a design decision
- ☐ Large-scale computation Optional
- ☐ Optimisation Optional

Thermal / Fluids / Energy Systems**Core areas**

Confirmed 1

Partial 0

Not yet 5

all 6

Confirmed so far

- ✓ Reflected in the design
bracket weight reduction capstone
- ✓ HVAC design Optional
bracket weight reduction capstone

Not yet confirmed

- ☐ Load calculation
- ☐ System layout
- ☐ Selecting equipment and conditions
- ☐ Analysis and calculation results
- ☐ Performance check
- ☐ Efficiency and energy Supporting
- ☐ Transient behaviour Supporting
- ☐ Operational control Supporting
- ☐ Test rig Supporting
- ☐ Renewables and power generation Optional

Automation / Robotics / Mechatronics**Core areas**

Confirmed 1

Partial 0

Not yet 5

all 6

Confirmed so far

- ✓ Mechanism and cell design
bracket weight reduction capstone

Not yet confirmed

- ☐ Deciding the scope of automation
- ☐ Actuator and sensor selection
- ☐ Control implementation
- ☐ Commissioning
- ☐ Automation result
- ☐ Safety design Supporting
- ☐ Vision and inspection Supporting
- ☐ Offline validation Supporting
- ☐ Return on investment Supporting
- ☐ A specific robot or PLC Optional
- ☐ Material handling automation Optional

Engineering Data / Digital Engineering

Core areas

Confirmed 1

Partial 0

Not yet 5

all 6

Confirmed so far

- ✓ Result validation
bracket weight reduction capstone
- ✓ Programming language Optional
bracket weight reduction capstone

Not yet confirmed

- ☐ Choosing what to automate
- ☐ Data structure design
- ☐ Building tools and scripts
- ☐ How it was used
- ☐ Someone else used it
- ☐ Usage instructions Supporting
- ☐ Change history Supporting
- ☐ Connecting to an existing system Supporting
- ☐ Visualisation Supporting
- ☐ Predictive model Optional

We do not turn the number of confirmed cells into a single We do not merge them into a number. Merging hides what the other cells were. An empty optional area is never held against you.

how the material was produced

- Interest, experience, decision ownership, work style and learning intent come from different questions and are read separately. We never merge them into one score.
- Experience is not part of your fit. It only changes how far your evidence is confirmed, differ.
- The evidence ladder is how deep one experience goes; evidence coverage is how much of the role's territory it covers. We never merge the two.
- A degree does not raise your evidence rung.

Limits

- Cognitive interviews and the pilot are still ahead, so we do not report a measurement error.
- These are not measures of ability or of your odds of being hired.
- The analysis below is written against the role itself. We did not ask for a target country, so Country-specific material such as licensing, salary, hiring practice and work authorisation is not included. Written without verified data, someone will believe it and act on it.
- The sixteen role families and their evidence areas hold for mechanical engineering only.

Version

Assessment	ME_V2_DECISION_2026
Result schema	3.0
Evidence map	1.0
Questions	68 of 68 answers
Generated	2026-10-06